Traditional Deep Learning & Transformers For Sexist Text Detection

Faezeh Sarlakifar, Ludovico Gorrieri, Alessandro Capialbi and **Giacomo Antonelli**
Master's Degree in Artificial Intelligence, University of Bologna
{ ludovico.gorrieri, alessandro.capialbi, giacomo.antonelli4, faezeh.sarlakifar }@studio.unibo.it

Abstract

Sexism detection in social media is a challenging NLP task due to contextual dependence and severe class imbalance. In this study, we address a four-class sexism detection task on tweets by evaluating LSTM architectures, as well as fine-tuned transformer models pretrained on hate speech and specific sexism-related data. To mitigate class imbalance, we apply data augmentation and a hierarchical classification strategy. The hierarchical approach achieved a macro F1 score of 0.55, demonstrating its effectiveness for this task.

1 Introduction

Early approaches to sexism detection relied on classical machine learning models or RNNs, typically combined with pretrained word embeddings. While effective for modeling local context, these methods are limited in handling long-range dependencies and implicit expressions of abuse (Badjatiya et al., 2017). More recently, transformer-based models pretrained on large-scale corpora have become the dominant approach for this task. Fine-tuned models such as BERT, RoBERTa, and especially social-media-specific variants like BERTweet have consistently achieved strong performance in sexism detection benchmarks and shared tasks (Nguyen et al., 2020).

2 System description

2.1 Data Preparation and Preprocessing

We prepared the dataset based on the instructions. To ensure data quality, we applied a strict label aggregation strategy that retained only samples with a clear majority vote and discarded tied cases, and then performed standard text preprocessing, including cleaning, lemmatization, and tokenization.

2.2 Data Augmentation

To address severe class imbalance, we applied random upsampling to augment minority classes, creating a balanced training set that improved generalization (hul, 2023).

2.3 Text Encoding

We constructed the vocabulary based on the training set tokens, employed pretrained GloVe embeddings that were trained on Twitter data, and initialized out-of-vocabulary (OOV) tokens from a uniform distribution (Pennington et al., 2014).

2.4 LSTM models

We trained a single-layer LSTM as a baseline and a stacked LSTM, both using pretrained GloVe embeddings.

2.5 Transformers

We fine-tuned `twitter-roberta-base-hate` (Barbieri et al., 2020) model, initializing a new classifier head for our four-class classification task. We didn't apply manual text cleaning, as our pre-trained transformer's tokenizer already handles it, and custom operations would affect model performance.

2.6 Bonus

We employed a hierarchical classification strategy, where a binary classifier first separated non-sexist from sexist text, followed by a multi-class classifier distinguishing among intents. See Figure 1 (Nakov et al., 2021).

We also evaluated a model pretrained specifically for sexism detection on tweets to assess the effect of using a task-specific model.

3 Experimental setup and results

We conducted comprehensive tuning using Optuna for the LSTM-based architectures and we used pretrained `glove.twitter.27B.50d` for LSTM and `glove.twitter.27B.100d` for stacked LSTM. Trainable embeddings resulted in better performance for the base LSTM while the opposite for stacked LSTM.

For transformer-based experiments, we fine-tuned `twitter-roberta-base-hate` (Transformer A), and, to assess the impact of task-specific pretrained models, we evaluated the `bertweet-large-sexism-detector` (Transformer B).

Table 1: Performance comparison of different models on test set (mean ± std)

Model	F1[1]	Prec[1]	Rec[1]
LSTM	0.35 ±0.04	0.37 ± 0.03	0.39±0.02
Stacked LSTM	0.39 ±0.04	0.40 ± 0.03	0.41± 0.03
Transformer A[2]	0.52 ± 0.01	0.51 ± 0.01	**0.58 ± 0.02**
Transformer B[3]	0.55 ± 0.02	**0.59 ± 0.01**	0.53 ± 0.03
Hierarchical model[4]	**0.55**	0.56	0.54

[1] Macro metrics
[2] Finetuned `twitter-roberta-base-hate`
[3] Finetuned `bertweet-large-sexism-detector`
[4] Finetuned `twitter-roberta-base-hate` for our hierarchical pipeline

Table 2: Performance comparison of different models on validation set

Model	F1	Prec	Rec
LSTM	0.34±0.05	0.37± 0.04	0.37±0.04
Stacked LSTM	0.40± 0.04	0.42± 0.03	0.43±0.03
Transformer A	0.52 ± 0.01	0.51 ± 0.02	**0.57 ± 0.01**
Transformer B	**0.59 ± 0.02**	**0.70 ± 0.04**	0.55 ± 0.04
Hierarchical model	0.55	0.58	0.54

Table 3: Performance comparison of `roberta-hate` on the test set, before and after train data augmentation

Augment	F1	Prec	Rec
No	0.46 ± 0.01	0.46 ± 0.02	0.49 ± 0.04
Yes	**0.52 ± 0.01**	**0.51 ± 0.01**	**0.58 ± 0.02**

4 Discussion

4.1 Quantitive results

Both LSTM and stacked LSTM performed poorly on the task with low F1 scores and strong overfitting slightly mitigated by augmentation. Table 1 shows that Transformer A achieves a macro F1 of 0.52 and the highest recall (0.58) on the test set, indicating better coverage of minority sexist classes. The task-specific Transformer B improves precision (0.59) and reaches a higher macro F1 of 0.55, suggesting that sexism-focused pretraining reduces false positives. This trend is consistent on the validation set (Table 2), where Transformer B attains the best macro F1 (0.59).

The hierarchical model matches the best model (Transformer B) performance on the test set (macro F1 = 0.55), showing that decomposing the task into binary and intent classification is consistently effective.

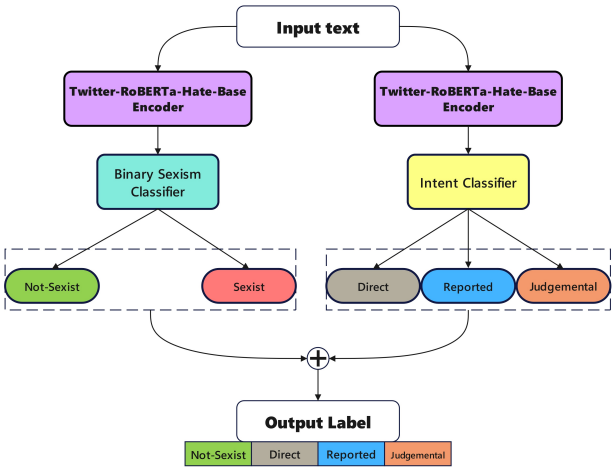

Figure 1: Hierarchical modeling inference pipeline

Finally, data augmentation improves model performance: for Transformer A, the macro F1 score increases to 0.52 (Table 1), with gains in precision and recall. Figure 2 further illustrates the effectiveness in improving generalization.

4.2 Error Analysis

Methods: Descriptive statistics (error distributions, entropy), Logistic Regression (Word count, Avg word length, Entropy, OOV), and SHAP feature analysis.
Base/Stacked LSTM: Weakest performance; overfits to majority class (*NOT_SEXIST*). Accuracy inversely correlates with entropy and length.
HateRoBERTa: Higher semantic capture. Augmented data reduces majority bias but struggles with *JUDGEMENTAL* vs. *REPORTED* distinction.
BERTweet Sexism: Superior semantic nuance but remains sensitive to specific keywords like "whore" or "murder" in SHAP analysis.
Hierarchical: Best at handling nuances but suffers from "gatekeeping" errors; if the initial binary sexism check fails, the sub-classification is blocked.

5 Conclusion

In this work, we compared LSTM-based and transformer-based models for sexism detection on tweets. Our experiments show that transformer models outperform LSTMs due to contextualized representations. Task-specific pretraining, data augmentation, and hierarchical classification further improved performance on imbalanced classes, highlighting the effectiveness of these techniques.

Despite these gains, limitations remain. Class imbalance and semantic overlap between sexism categories continue to cause misclassifications.

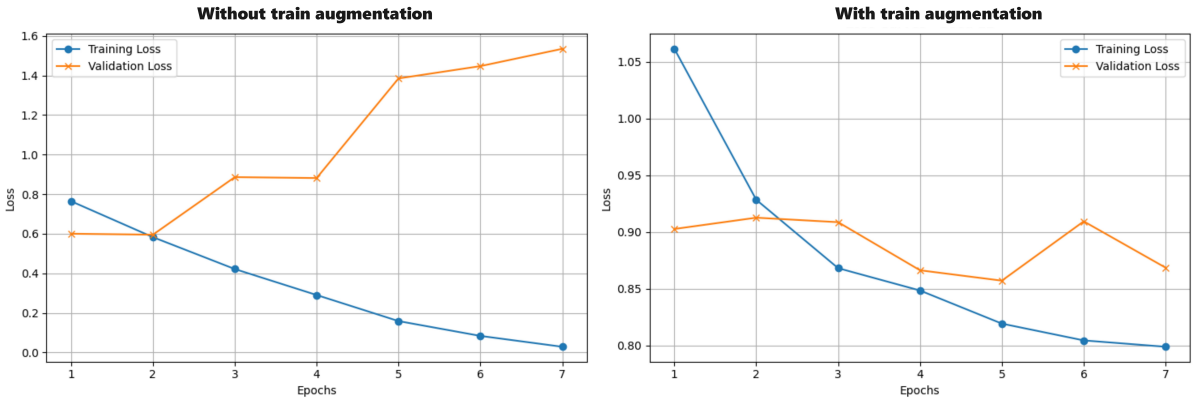

Figure 2: Learning curves of our main transformer model before and after augmentation

References

2023. HULAT at SemEval-2023 Task 10: Data augmentation for pre-trained transformers applied to the detection of sexism in social media. In *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023)*, Toronto, Canada. Association for Computational Linguistics.

Pinkesh Badjatiya et al. 2017. Deep learning for hate speech detection. In *WWW Companion*.

Francesco Barbieri et al. 2020. Tweeteval: Unified benchmark for tweet classification. In *Findings of EMNLP*.

Preslav Nakov et al. 2021. Semeval-2021 task 5: Toxic spans detection. In *SemEval*.

Dat Quoc Nguyen et al. 2020. Bertweet: A pre-trained language model for english tweets. In *EMNLP*.

Jeffrey Pennington, Richard Socher, and Christopher Manning. 2014. Glove: Global vectors for word representation. In *EMNLP*.